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**Efficient Removal of Tetracycline from Aqueous Media Using a Novel
Zr-MOF/CuCo₂O₄ Nanocomposite: Synthesis, Characterization, and Mechanistic Insights**

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Abstract

In this research, a new Zr-MOF/CuCo₂O₄ nanocomposite was successfully synthesized and evaluated for the efficient removal of tetracycline (TC) from aqueous solutions. The composite displayed significant adsorption capacity, fast kinetics, and outstanding reusability. Analytical techniques including FT-IR, XRD, FE-SEM, BET, TGA, DSC, NMR and zeta potential confirmed the formation and stability of the nanocomposite. Optimization of the removal conditions was performed using Response Surface Methodology (RSM) with a Central Composite Design (CCD), achieving >99% TC removal under optimum conditions (pH 7.80, 17.50 mgL⁻¹ TC solution, 7.0 mg adsorbent, 46 min). Thermodynamic studies indicated a spontaneous and endothermic process, while Kinetic analysis indicated that the adsorption process followed a pseudo-second-order model. The nanocomposite maintained high stability and efficiency over seven reuse cycles. Real-sample analysis using seawater and pharmaceutical wastewater confirmed its strong potential for practical environmental applications. These findings highlight Zr-MOF/CuCo₂O₄ as a promising adsorbent for removing antibiotics from contaminated water, contributing to sustainable water purification and pollution control.

Keywords Zr-MOF/CuCo₂O₄ nanocomposite, tetracycline removal, adsorption, response surface methodology, environmental remediation

Introduction

Medicines are essential components of modern life, serving as key agents in treating human, animal, and plant diseases. Among them, antibiotics as a major class of pharmaceutical compounds have attracted significant attention as emerging organic micropollutants due to their environmental persistence¹. These compounds often require targeted chemical pathways for removal, as they remain stable under typical wastewater treatment processes.

Antibiotics, widely used to control bacterial infections, can enter aquatic environments through various pathways, including industrial discharges². Once released, they contaminate surface and groundwater systems and contribute to the proliferation of antibiotic-resistant bacteria, posing a serious global health risk. Consequently, the development of efficient antibiotic removal strategies is a critical area of environmental research³.

Tetracyclines (TCs) represent a significant class of antibiotics frequently detected in wastewater and natural water bodies⁴. These bacteriostatic, broad-spectrum antibiotics are effective against a wide range of bacteria and some intracellular pathogens⁵. TCs are either produced naturally via microbial fermentation or synthesized chemically⁶. Structurally, they are characterized by a four-ring backbone with variable substituent groups⁷, inhibiting protein synthesis by binding to bacterial ribosomal subunits. Common examples include tetracycline, oxytetracycline, chlorotetracycline, and doxycycline⁸, which exhibit similar antimicrobial profiles but differ in their pharmacological and physicochemical properties⁹.

Metal–organic frameworks (MOFs) have attracted considerable interest due to their exceptionally high surface area and tunable functionality^{10,11}. These materials consist of metal clusters coordinated to organic ligands, forming one-, two-, or three-dimensional networks^{12–14}. Owing to their adjustable pore size and chemical stability, MOFs are increasingly employed for adsorption, catalysis, sensing, and environmental remediation^{15,16}. Zr-based MOFs (Zr-MOFs), in particular, exhibit high chemical stability and superior

adsorption performance, attributed to their abundant Lewis acid sites and robust coordination frameworks¹⁷⁻¹⁹.

Furthermore, luminescent and hybrid MOFs have been explored for pollutant detection owing to their high sensitivity and selectivity^{20,21}. The integration of detection and adsorption capabilities enables the development of multifunctional materials that can simultaneously capture and sense hazardous contaminants^{22,23}. Various studies have reported the removal and degradation of pollutants using nanocomposites²⁴⁻²⁶.

In this research, a novel Zr-MOF/CuCo₂O₄ nanocomposite was synthesized and investigated as a multifunctional material for the removal of tetracycline from water. The nanocomposite exhibited high efficiency, stability, and reusability. A suite of advanced characterization techniques—including FT-IR, XRD, FE-SEM/EDX mapping, BET surface area analysis, zeta potential, TGA, and NMR—was employed to elucidate its structural, morphological, and surface properties. The adsorption conditions were optimized using Response Surface Methodology (RSM) to achieve maximum tetracycline removal.

Experimental Section

Materials and Instrumentation

All chemicals were of analytical grade and used without further purification. Zirconium oxychloride octahydrate (ZrOCl₂·8H₂O, 98%), benzene-1,4-dicarboxylic acid (H₂BDC, >99%), acetic acid (>99.5%), and N,N-dimethylformamide (DMF, 99.8%) were purchased from Merck (Darmstadt, Germany). Copper(II) nitrate hexahydrate (Cu(NO₃)₂·6H₂O, 99%), cobalt(II) nitrate hexahydrate (Co(NO₃)₂·6H₂O, 99.9%), nickel(II) nitrate hexahydrate (Ni(NO₃)₂·6H₂O, >99.9%), calcium nitrate tetrahydrate (Ca(NO₃)₂·4H₂O, 99.9%), magnesium chloride hexahydrate (MgCl₂·6H₂O, 98%), and analytical-grade pharmaceuticals—including tetracycline (≥98.5%), ofloxacin (99.9%), sulfamethoxazole (99%),

trimethoprim (99.9%), cefazolin (95–102%), azithromycin (99%), and cefixime (98%)—were obtained from Merck. Caffeine (98.5%), glucose (>99%), hydrochloric acid (HCl, 37%), and sodium hydroxide (NaOH, $\geq 99\%$) were also used.

Fourier-transform infrared (FT-IR) spectra were recorded using a (Bruker Tensor 27) spectrometer (Germany) to identify functional groups. X-ray diffraction (XRD) patterns were obtained using a (PANalytical X'Pert PRO) diffractometer (Herzog, Germany) to confirm crystallinity. Morphology and microstructure were analyzed by FE-SEM (MIRA3 TESCAN, Czech Republic) equipped with energy-dispersive X-ray spectroscopy (EDX) and elemental mapping. Thermal stability was examined via (TGA/DSC) analysis (NETZSCH STA 409 PC/PG, Germany) under an air atmosphere at $10\text{ }^{\circ}\text{C min}^{-1}$.

^1H and ^{13}C nuclear magnetic resonance (NMR) spectra were recorded on a (Bruker AV 300 MHz Avance-III) spectrometer using DMSO- d_6 (for benzene dicarboxylic acid) and DMSO- d_6 /DCl (for Zr-MOF/ CuCo_2O_4) as solvents. Surface area and pore characteristics were analyzed using (BET and BJH) methods (BELSORP MINI II, BEL Japan). Dynamic light scattering (DLS; Bruker, Germany) determined particle size distribution and surface charge (zeta potential). TC concentration was monitored using a (Varian Cary 50 UV–Vis) spectrophotometer (Melbourne, Australia) at $\lambda_{\text{max}} = 365\text{ nm}$.

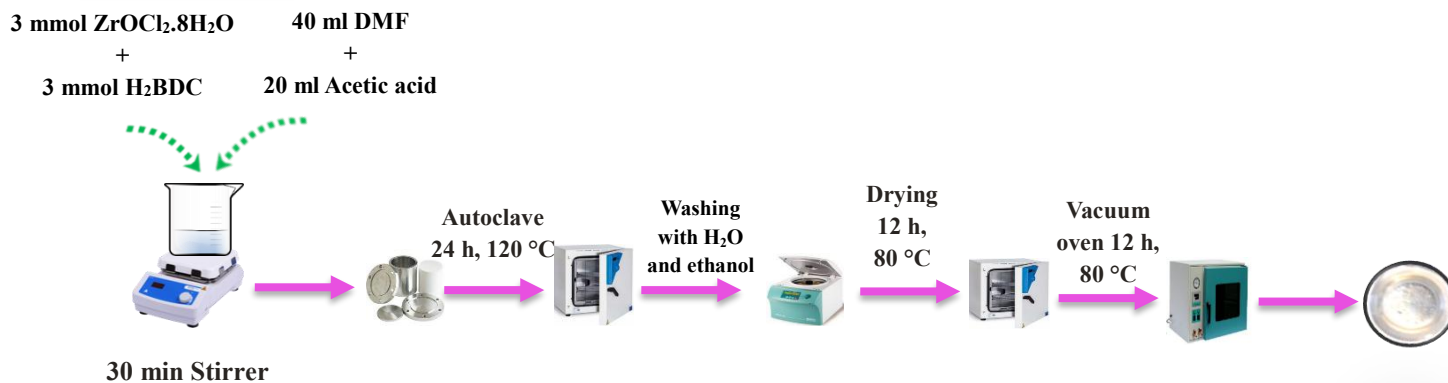
Synthesis of Zr-MOF

Zr-MOF was synthesized via a modified solvothermal method adapted from previous literature^{27,28}. Briefly, equimolar amounts of $\text{ZrOCl}_2 \cdot 8\text{H}_2\text{O}$ and H_2BDC (3 mmol each) were dissolved in 40 mL DMF within an autoclave. Acetic acid (20 mL) was added as a modulator. The sealed vessel was heated at $120\text{ }^{\circ}\text{C}$ for 24 h. The resulting precipitate was collected by centrifugation (4200 rpm, 5 min), washed repeatedly with methanol and water, and dried at $80\text{ }^{\circ}\text{C}$ for 12 h, followed by vacuum drying at the same temperature²⁹.

Synthesis of Zr-MOF/ CuCo_2O_4 Nanocomposite

To prepare the nanocomposite, CuCo_2O_4 nanoparticles were synthesized in situ and incorporated into the Zr-MOF structure using a solution-based method. Typically, 1.0 g of Zr-MOF was dispersed in 100 mL ethanol under ultrasonication for 5 min. A stoichiometric aqueous mixture of $\text{Cu}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (1:2 molar ratio) along with urea was added to the Zr-MOF suspension. Urea acts as a precursor for carbonate ions, which, upon heating, facilitates the nucleation and growth of metal oxide nanoparticles. The mixture was stirred at 60 °C for 1 h, allowing for the in situ formation and deposition of CuCo_2O_4 onto the Zr-MOF surface. Subsequently, the solvent was evaporated, and the resulting solid was dried at 80 °C overnight (Fig. 1). For comparison, pure CuCo_2O_4 nanoparticles were synthesized according to a reported procedure³⁰.

Synthesis of Zr-MOF



Synthesis of Zr-MOF/ CuCo_2O_4

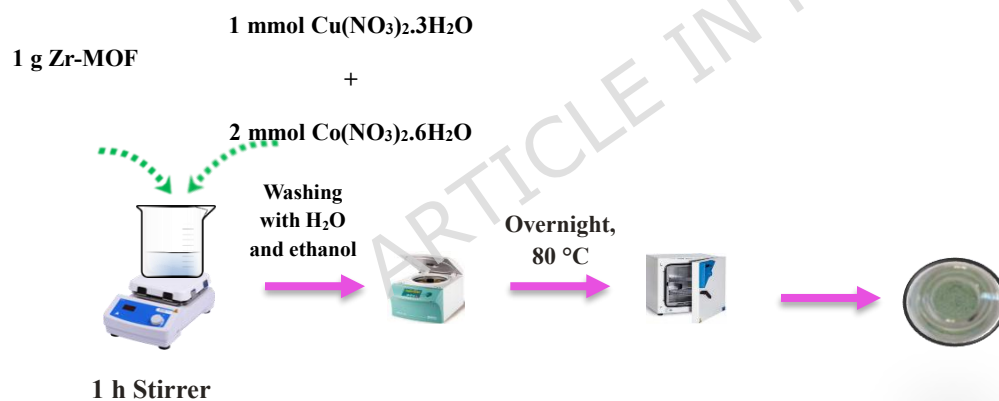


Fig. 1. Steps for construction of samples.

Adsorption Experiments

Batch adsorption experiments were performed according to the conditions in the experimental design, using a volume of 10 mL. The suspension was stirred for 46 min, followed by centrifugation

(4000 rpm, 3 min). The supernatant's absorbance was measured at 365 nm to determine the residual TC concentration.

The TC removal efficiency (R%) was calculated using Eq. (1):

$$R\% = \frac{A_0 - A_t}{A_0} \times 100 \quad (1)$$

where A_0 and A_t are the initial and time-dependent absorbances of the TC solution at 365 nm³¹.

Results and Discussion

Characterization

The synthesized materials were systematically characterized using FT-IR, XRD, FE-SEM/EDX mapping, BET, TGA, DSC, NMR and zeta potential analyses to elucidate their composition, structure, and physicochemical properties.

FT-IR spectra (Fig. 2) confirmed the formation of characteristic bonds in Zr-MOF, CuCo₂O₄, and the Zr-MOF/CuCo₂O₄ nanocomposite. The Zr-MOF spectrum exhibited peaks at 660 and 770 cm⁻¹ (Zr–O), 1385 cm⁻¹ (Zr–OH and C=C), 1500–1650 cm⁻¹ (C–O–Zr), 1658 cm⁻¹ (C=O), and 3200–3600 cm⁻¹ (–OH stretching), indicating the presence of organic linkers and coordinated metal centers³². The (Fig. 2a) is taken from the authors' previous published paper³³. CuCo₂O₄ displayed peaks at 3211 and 1613 cm⁻¹ corresponding to H–O–H vibrations of adsorbed water molecules, and at 722.9 and 1107 cm⁻¹, attributed to Co–O and Cu–O stretching in tetrahedral and octahedral environments³⁴. In the Zr-MOF/CuCo₂O₄

spectrum, combined features at 730 cm^{-1} (Zr–O/Cu–O), 1385 cm^{-1} (Zr–OH), 1550 cm^{-1} (C–O–Zr), and 1200 cm^{-1} (Co–O) confirmed successful hybridization.

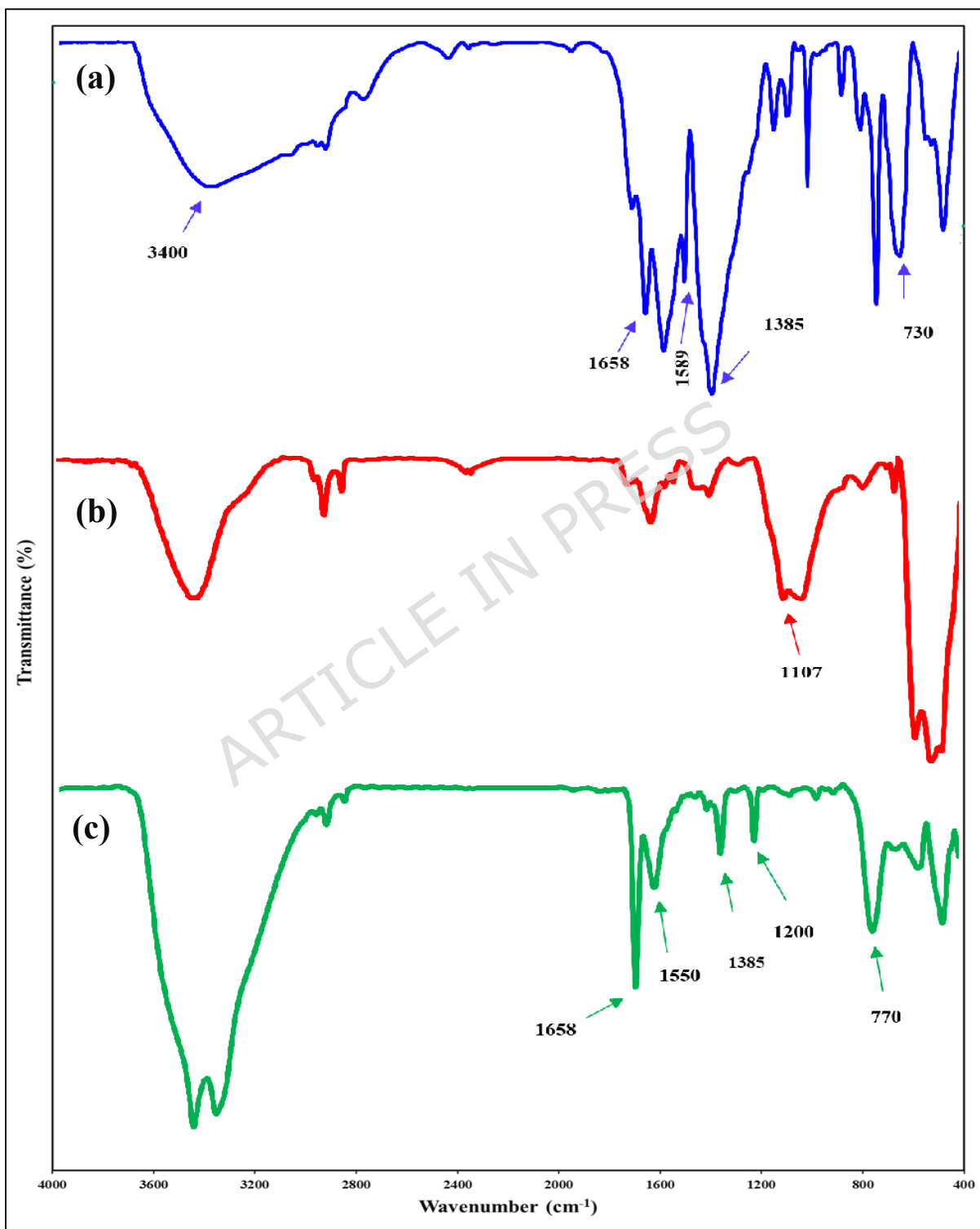


Fig. 2. FT-IR spectra of (a) Zr-MOF³³, (b) CuCo₂O₄ and (c) Zr-MOF/CuCo₂O₄ nanocomposite.

XRD analysis (Fig. 3) demonstrated that Zr-MOF exhibited diffraction peaks corresponding to its hexagonal phase (JCPDS #84-0127)³⁵, while CuCo₂O₄ displayed peaks at 2θ values of 18.9°, 31.3°, 36.9°, 38.6°, 44.9°, 59.5°, and 65.4°, corresponding to (111), (220), (311), (222), (400), (511), and (440) planes of its cubic spinel phase (JCPDS #01-1155)³⁶. The (Fig. 3a) was used from the authors' previous published paper³³. The diffraction pattern of the Zr-MOF/CuCo₂O₄ nanoadsorbent revealed features of both structures, confirming successful integration.

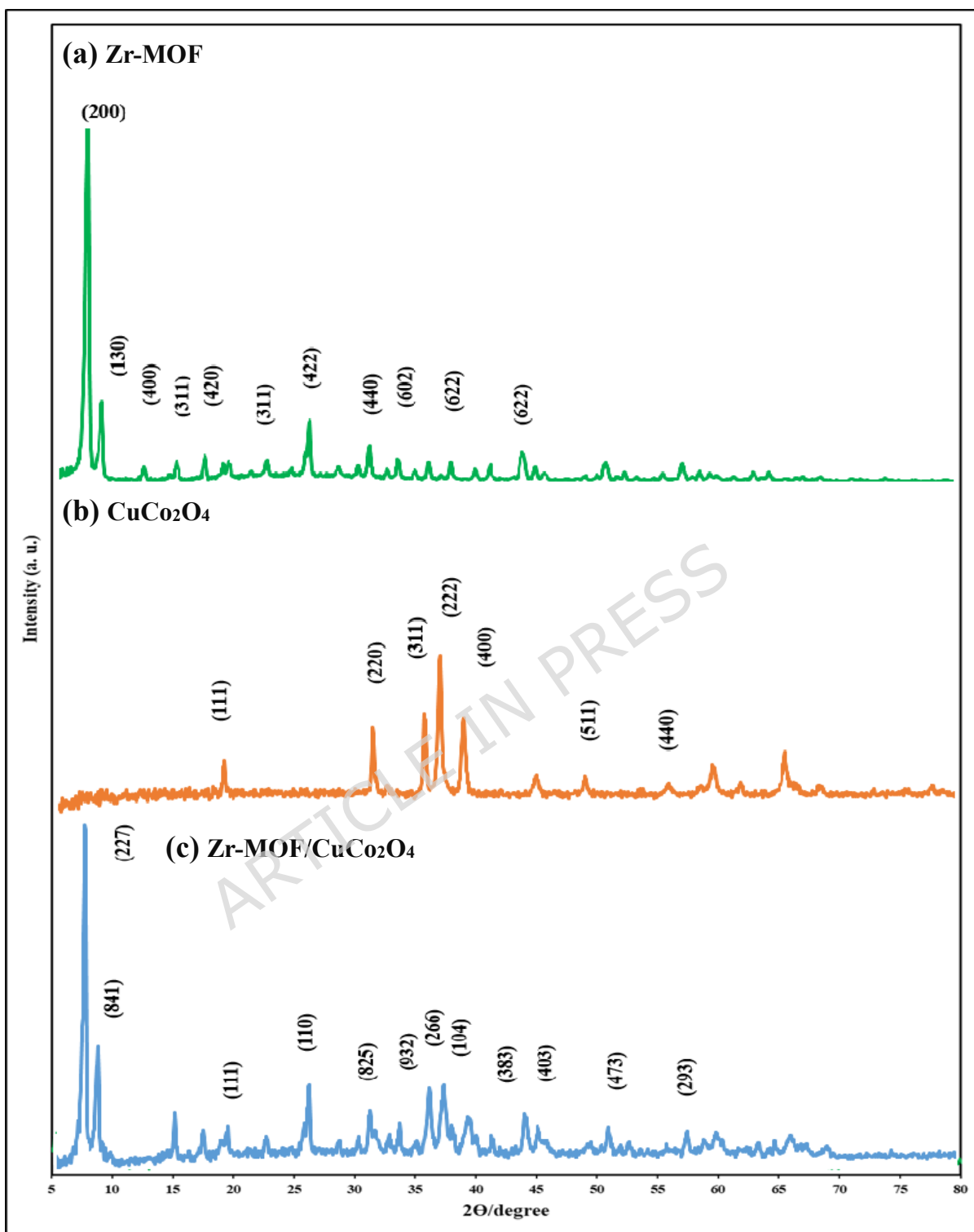


Fig. 3. XRD of (a) Zr-MOF³³, (b) CuCo_2O_4 and (c) Zr-MOF/ CuCo_2O_4 nanocomposite.

FE-SEM images (Fig. 4) showed that Zr-MOF crystals possessed a polyhedral morphology, while CuCo_2O_4 nanoparticles exhibited a near-spherical structure. The (Fig. 4a) in another magnification was used from the authors' previous published paper³³.

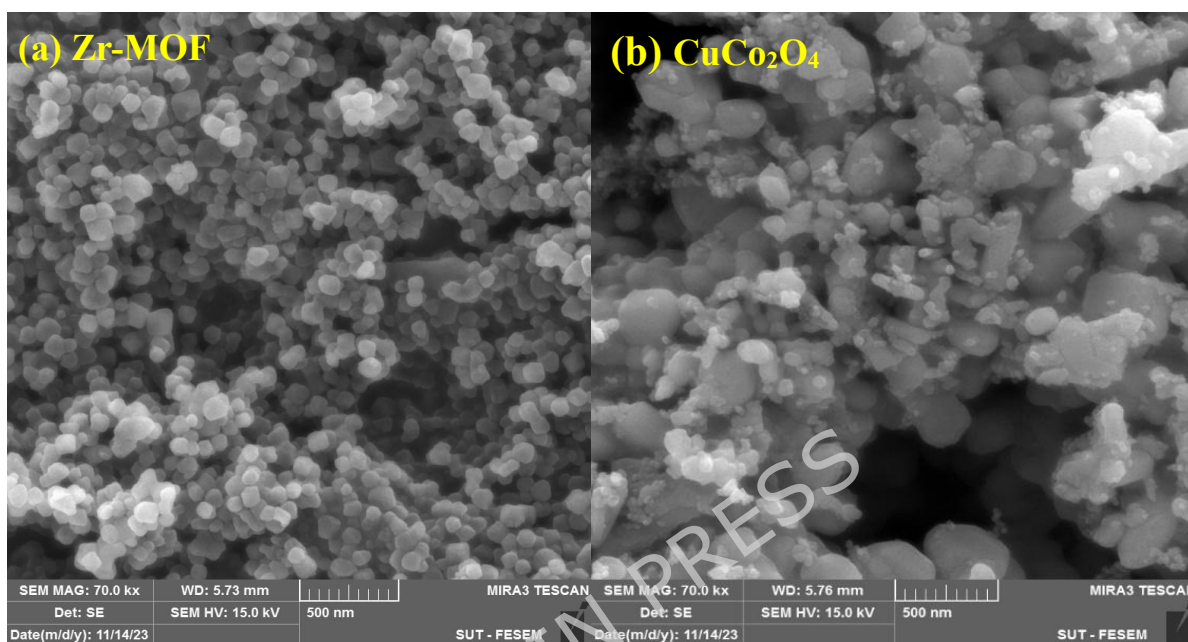
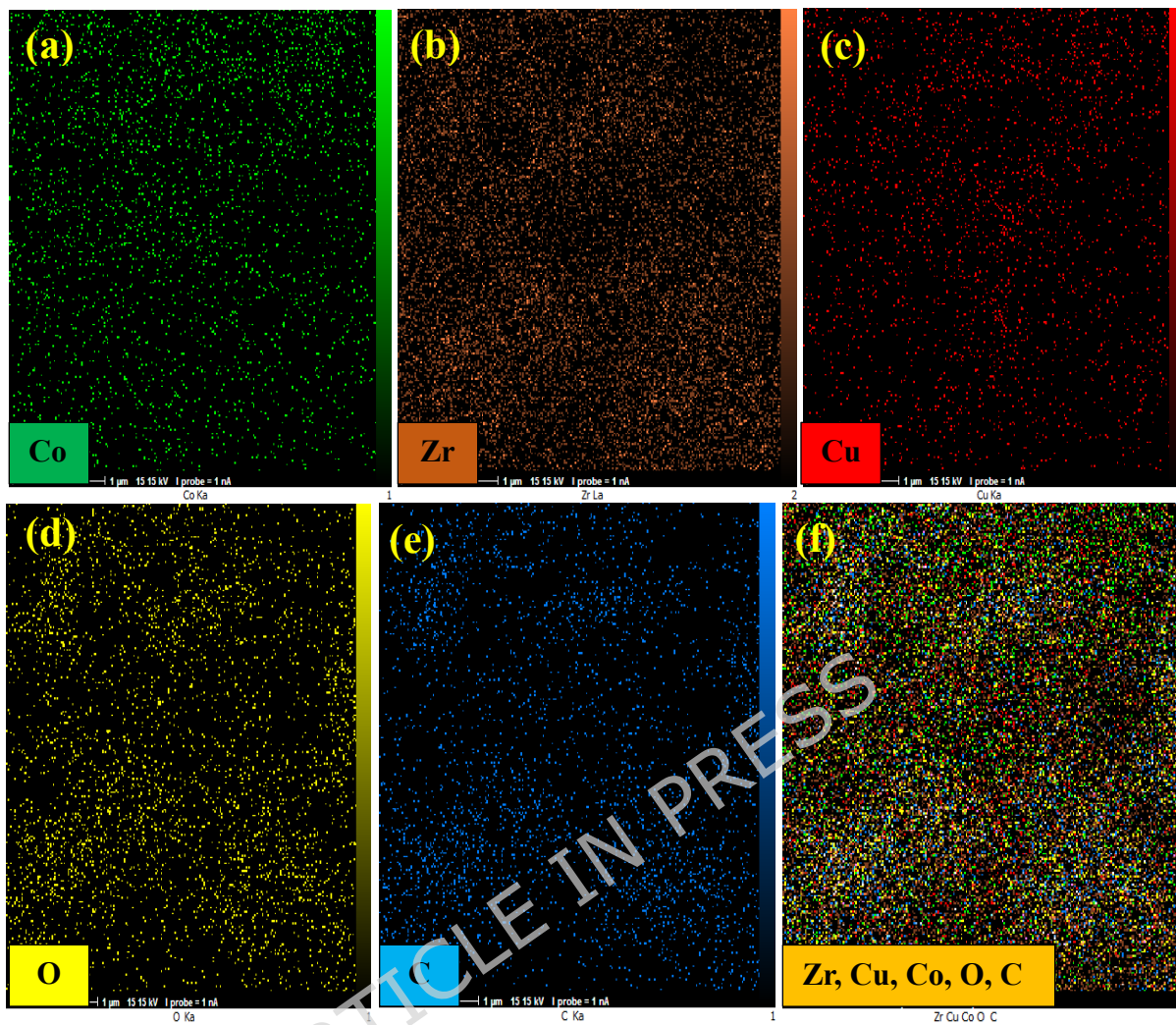


Fig. 4. FE-SEM images of (a) Zr-MOF³³, (b) CuCo_2O_4 and (c) Zr-MOF/ CuCo_2O_4 nanocomposite.

Elemental mapping (Fig. 5a–f) confirmed homogeneous dispersion of Zr, C, O, Cu, and Co elements. The nanocomposite displayed uniform distribution of both phases. (Fig. 5g) FESEM image of the nanocomposite from which the mapping was taken, while EDX spectra (Fig. 5h) verified the purity of the product. EDS analysis nanocomposite shows Zr (46.16%), C (10.40%), O (20.38%), Cu (7.33%), and Co (15.73%) elements. The mean size was measured to be 67.82 nm for Zr-MOF, 40.35 nm for CuCo_2O_4 , and 43.95 nm for the composite.



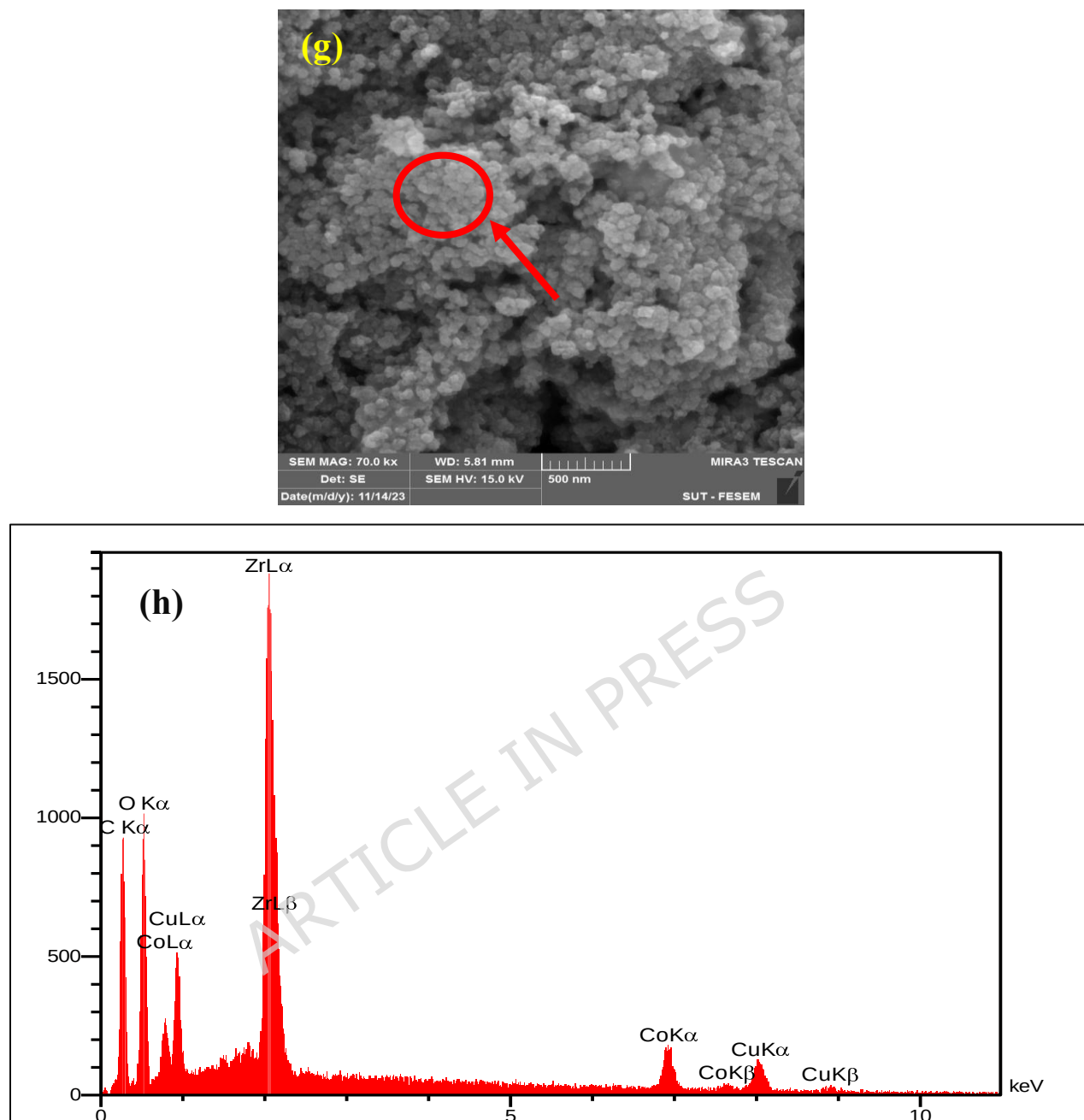
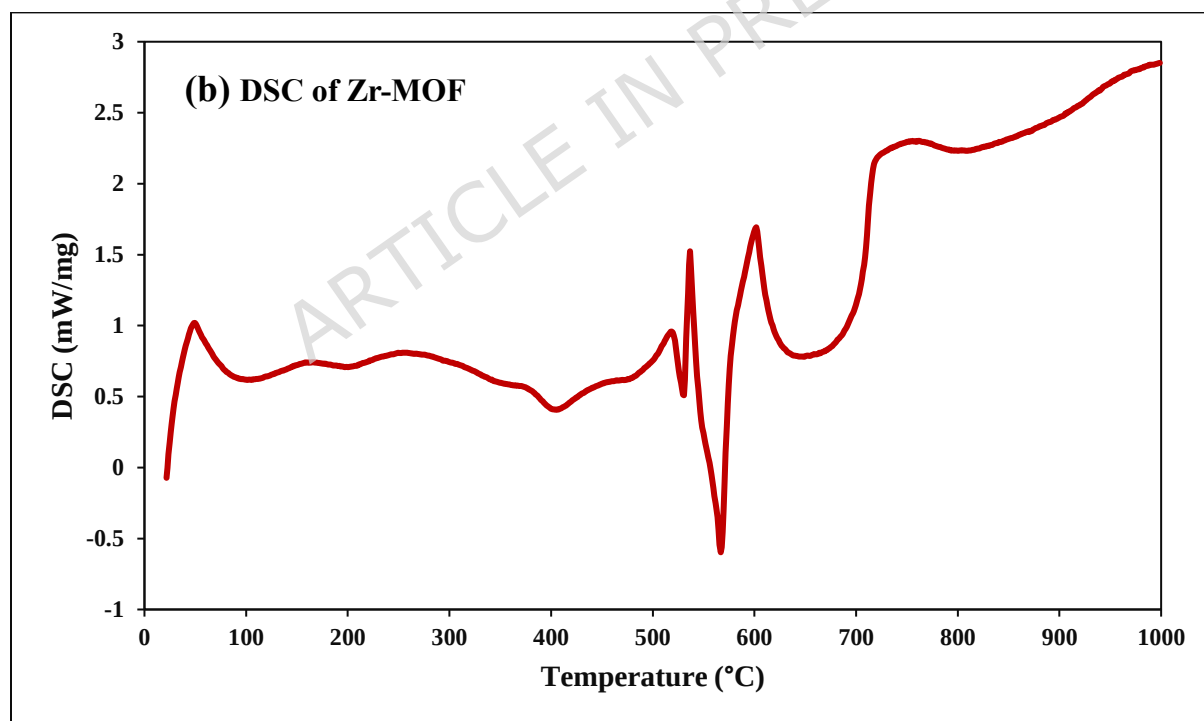
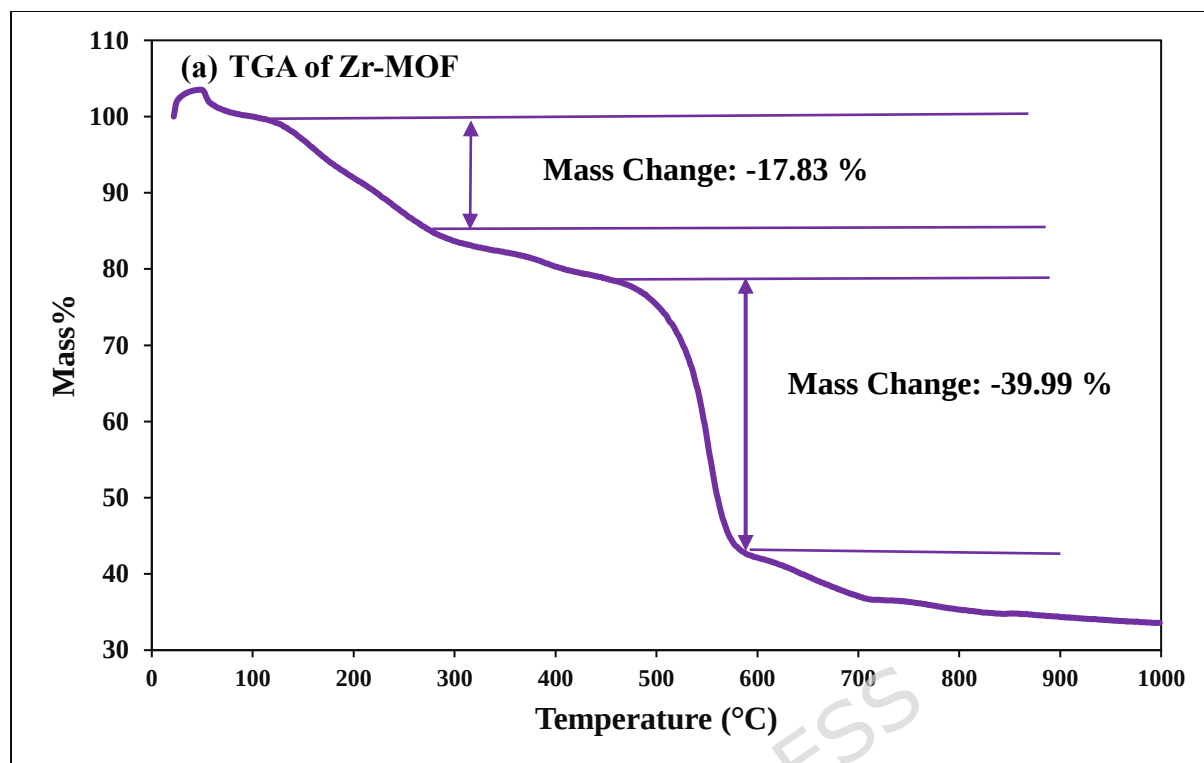


Fig. 5. (a–f) Nanoadsorbent mapping analysis, (g) FE-SEM image of the spot of which mapping was taken (corresponding to nanoadsorbent) and (h) EDX analysis.

TGA and DSC analyses (Fig. 6) were used to assess the thermal stability of the materials. The Zr-MOF showed two major weight-loss stages: the first (25–293 °C) corresponding to solvent evaporation and removal of guest molecules, and the second (414–600 °C) indicating framework decomposition. The

DSC curve showed an exothermic peak at ~ 580 °C, confirming stability up to this temperature. In contrast, the nanocomposite exhibited initial mass loss between 25–282 °C (due to desorption of surface-bound water and solvents) and a second loss between 393–800 °C, suggesting decomposition of the hybrid structure^{37,38}.

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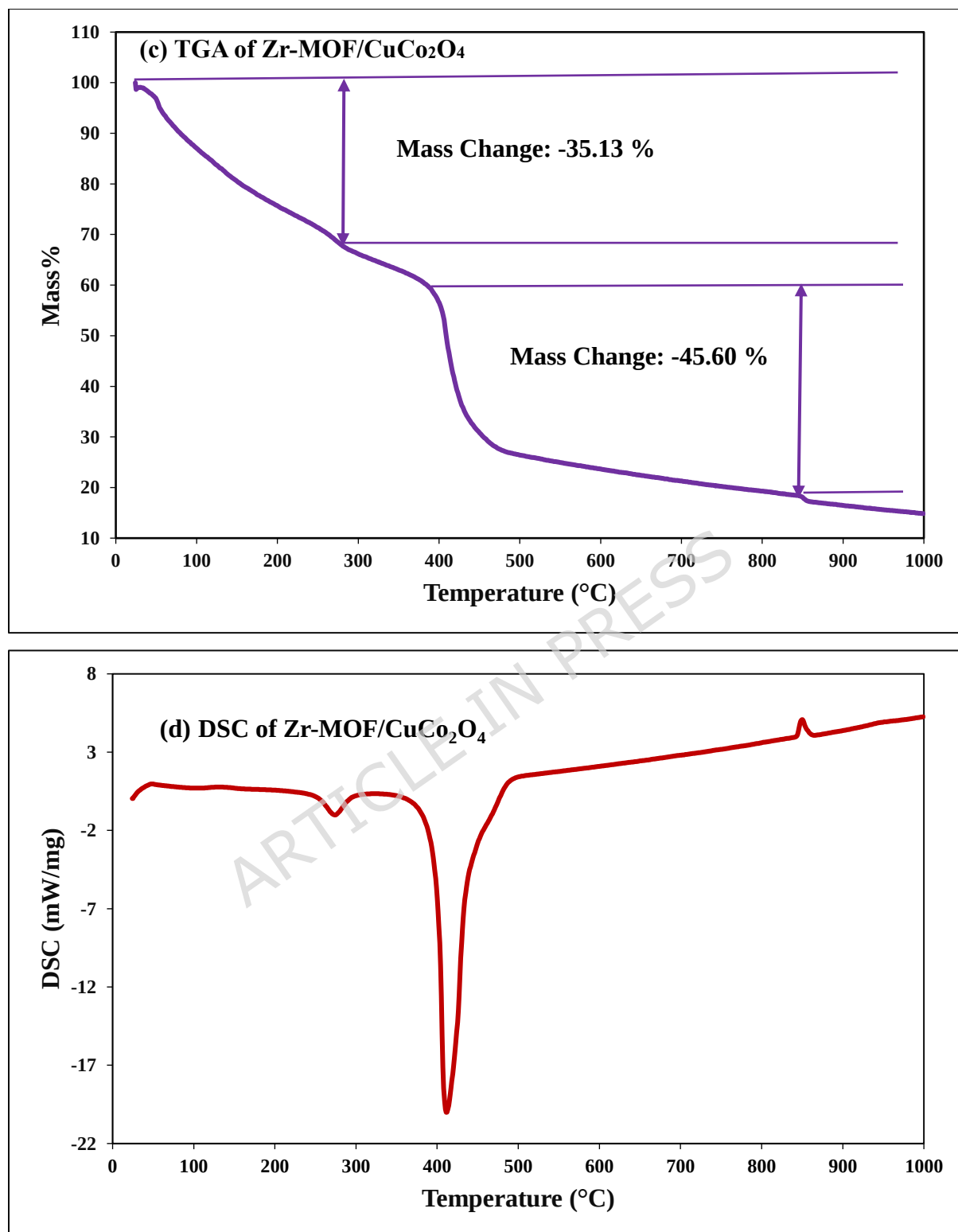
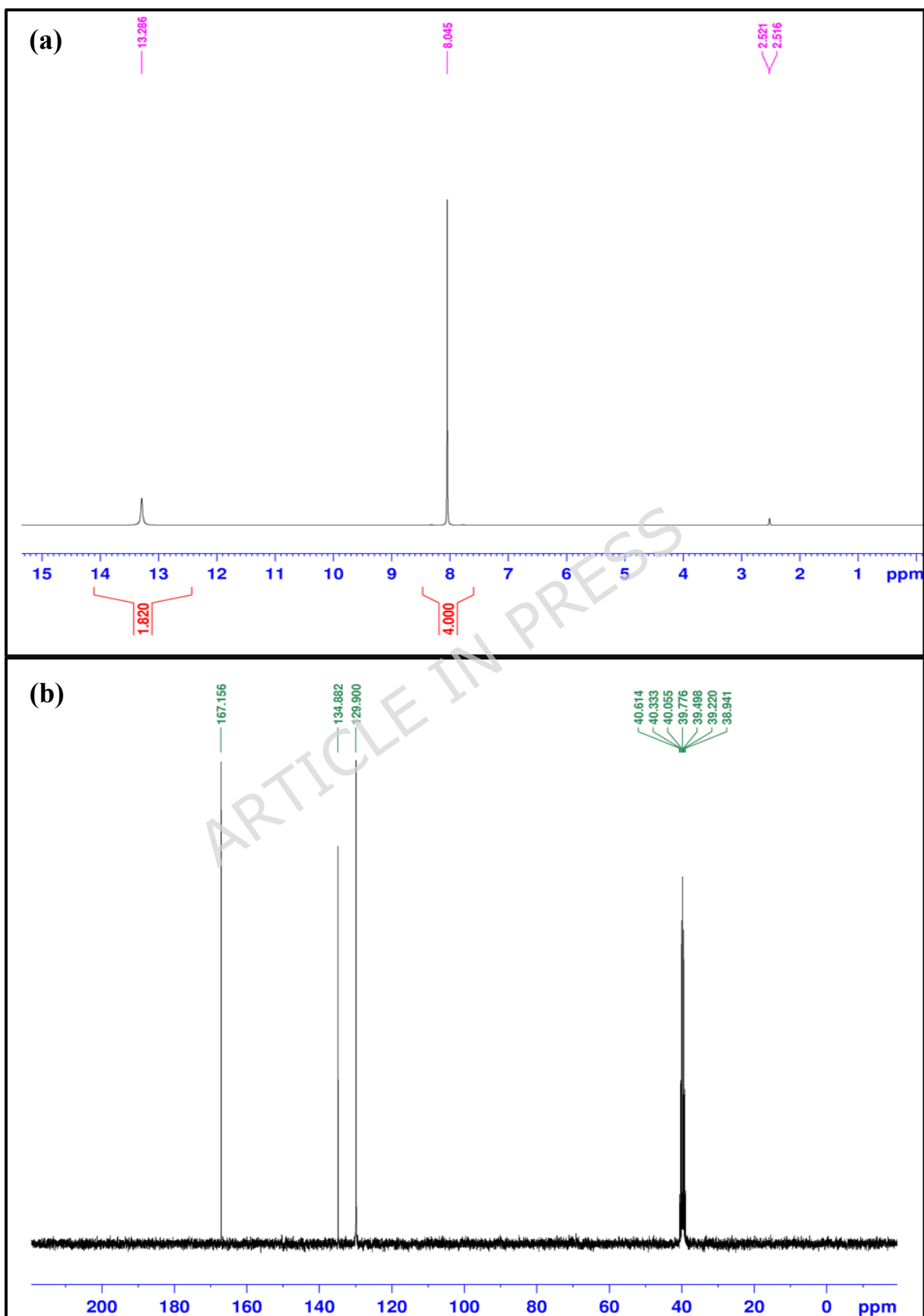


Fig. 6. Curves of TGA and DSC of samples in 25-1000 °C.

^1H and ^{13}C NMR spectra (Fig. 7) further validated structural integrity. The characteristic aromatic proton signals in ^1H NMR for benzene dicarboxylic acid appeared at 8.04 ppm (aromatic H) and 13.28 ppm (acidic H), whereas ^{13}C NMR showed resonances at 129.9 ppm, 134.8 ppm (aromatic C), and 167.1 ppm (carbonyl C). In the Zr-MOF/ CuCo_2O_4 composite, aromatic signals were observed at 7.95 ppm, while acidic hydrogen peaks disappeared due to coordination with metal centers.



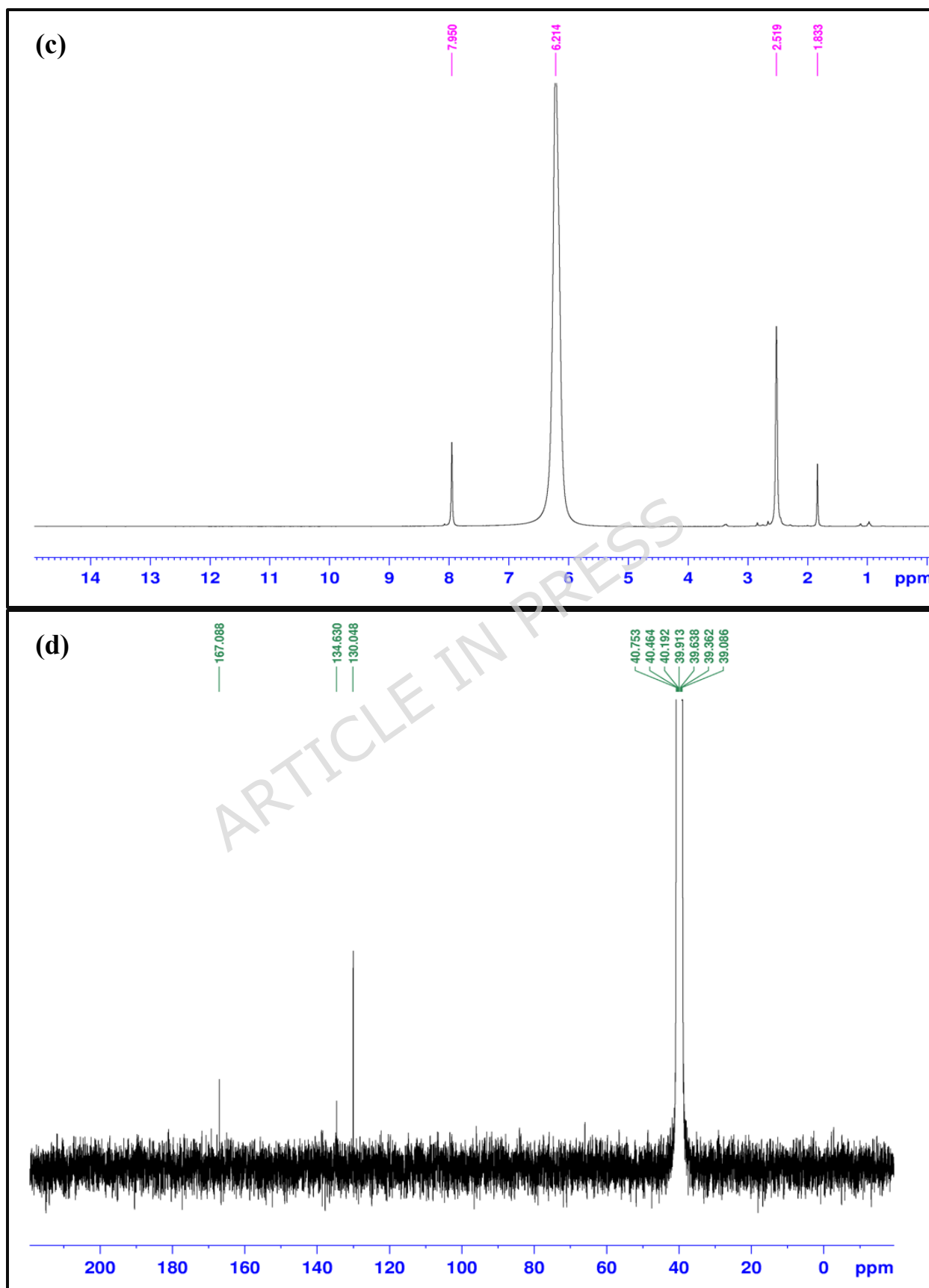


Fig. 7. a) ^1H -NMR spectrum of benzene dicarboxylic acid, b) ^{13}C -NMR spectrum of benzene dicarboxylic acid, c) ^1H -NMR spectrum of Zr-MOF/ CuCo_2O_4 , and d) ^{13}C -NMR spectrum of Zr-MOF/ CuCo_2O_4 nanocomposite.

BET isotherm analysis (Fig. 8, Table 1) revealed distinct adsorption behaviors for the individual components and the resulting nanocomposite. The pure Zr-MOF exhibited a Type II isotherm, characteristic of macroporous materials with relatively high surface area and pore volume (Table 1), indicating good accessibility for adsorbate molecules. In contrast, the pure CuCo_2O_4 nanoparticles displayed a Type III isotherm, often associated with weaker adsorbate-adsorbent interactions and potentially smaller, less accessible pores. Upon forming the Zr-MOF/ CuCo_2O_4 nanocomposite, a significant decrease in both BET surface area (from $991.01 \text{ m}^2\text{g}^{-1}$ for Zr-MOF to $102.02 \text{ m}^2\text{g}^{-1}$) and pore volume (from $227.69 \text{ cm}^3\text{g}^{-1}$ to $63.958 \text{ cm}^3\text{g}^{-1}$) was observed (Table 1). This reduction strongly suggests the successful incorporation and deposition of CuCo_2O_4 nanoparticles within the Zr-MOF structure, partially blocking some of the larger pores.

BJH analysis (Fig. 9) confirmed a microporous structure with average pore sizes of 1.21 nm (Zr-MOF) and 1.84 nm (Zr-MOF/ CuCo_2O_4).

Parameter (Unit)	Zr-MOF	CuCo_2O_4	Zr-MOF/ CuCo_2O_4
$V_a(p_0-p)$ (amount of nitrogen adsorbed at given p/p_0) (cm^3)	19256	56.684	14.695
V_m (monolayer volume) ($\text{cm}^3(\text{STP}).\text{g}^{-1}$)	227.69	0.8475	63.958
$a_{s,\text{BET}}$ (Surface Area) ($\text{m}^2.\text{g}^{-1}$)	991.01	3.6887	102.02
C (BET constant) (-)	773.1	17.973	10. 403
Total pore volume ($p/p_0=0.990$) (p/p_0 =relative pressure of nitrogen) ($\text{cm}^3.\text{g}^{-1}$)	0.8675	0.015395	0.7923

Table 1. BET surface area analysis of samples.

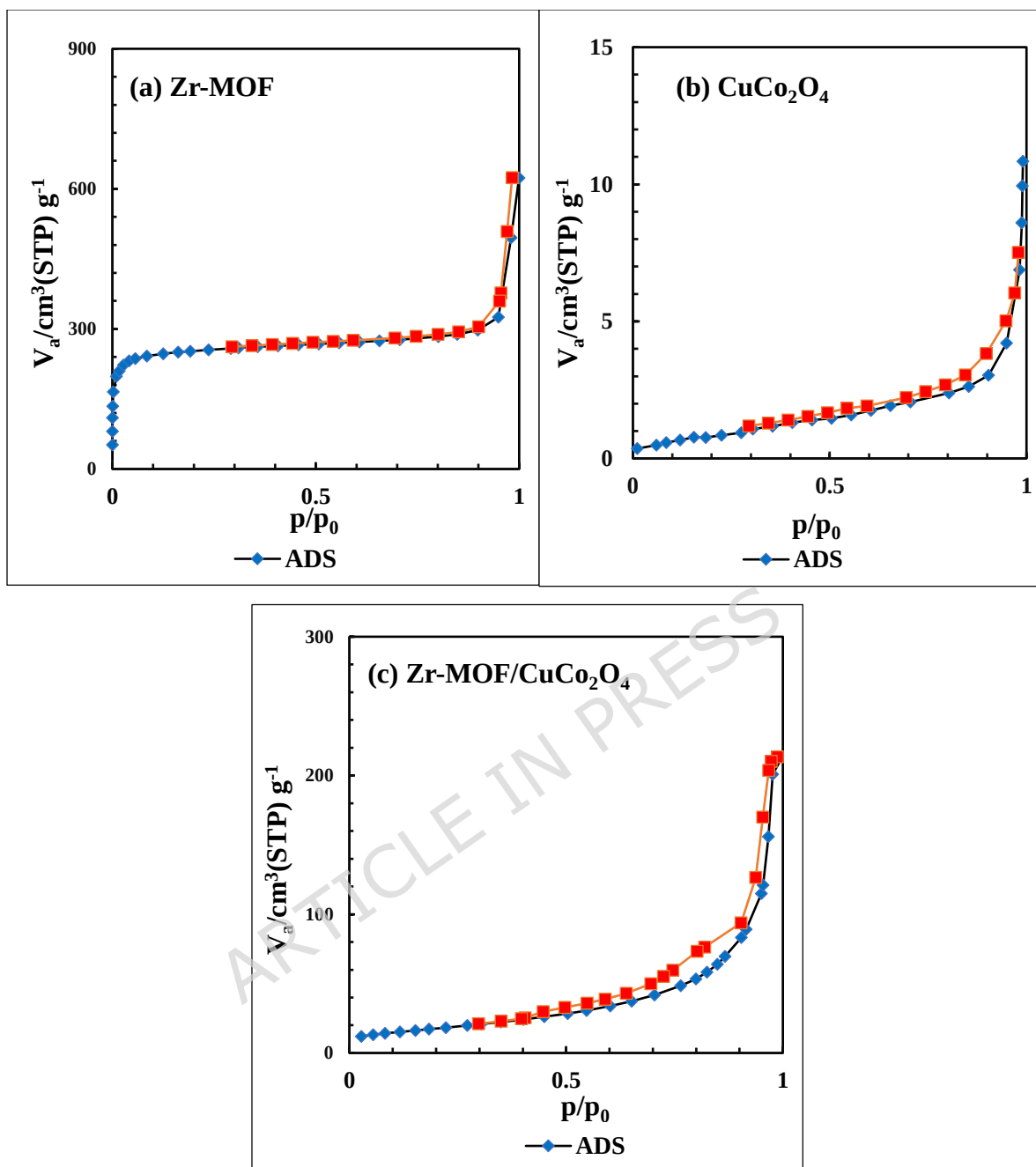


Fig. 8. N₂ adsorption-desorption isotherms of isotherms of (a) Zr-MOF, (b) CuCo₂O₄ and (c) Zr-MOF/CuCo₂O₄ nanocomposite.

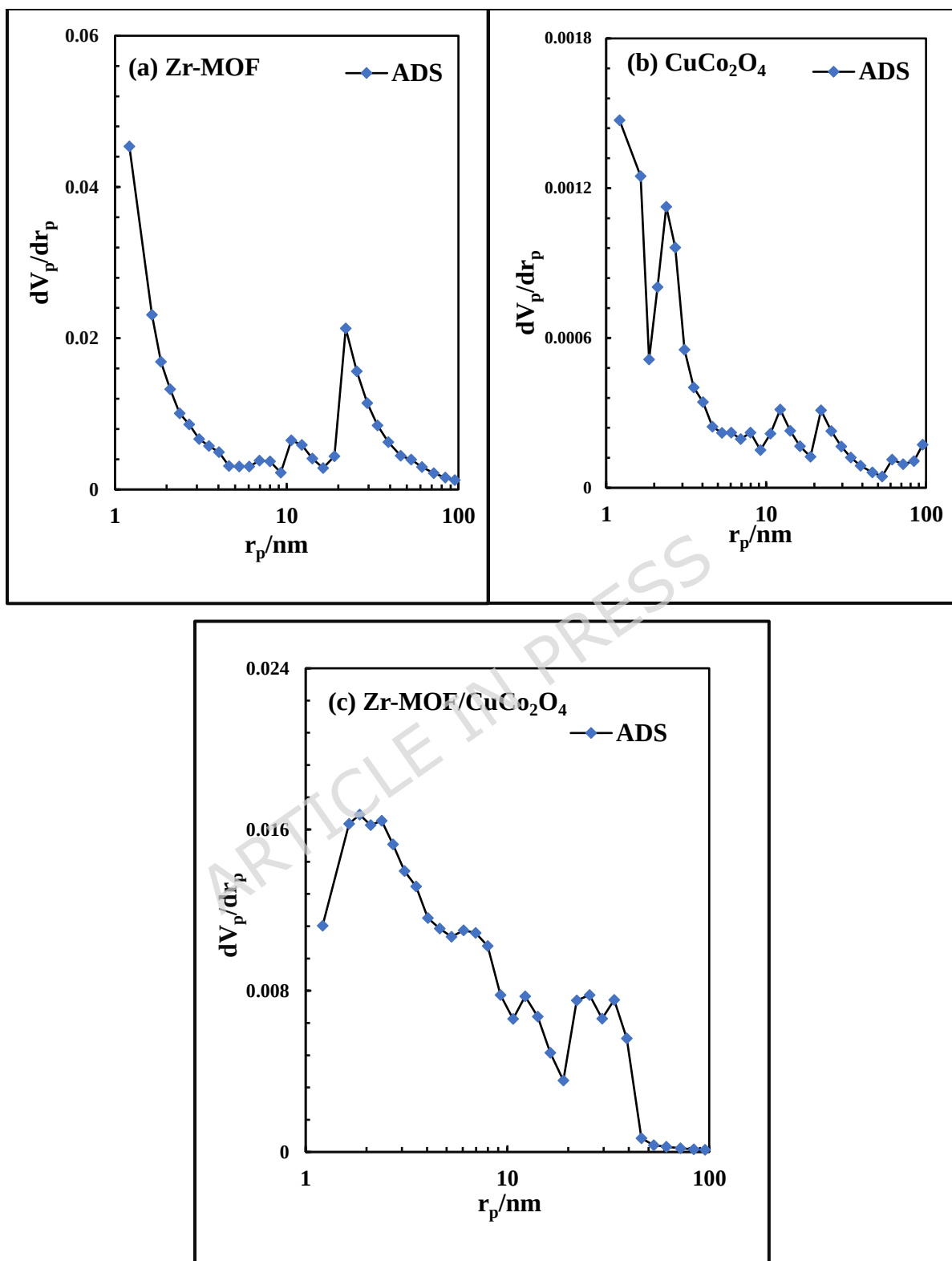


Fig. 9. BJH-plot curves of (a) Zr-MOF, (b) $CuCo_2O_4$ and (c) Zr-MOF/ $CuCo_2O_4$ nanocomposite.

The point of zero charge was obtained as 5.15. Zeta potential analysis (Fig. 10) revealed surface charge values at pH 2 to 10 (from +42.40 to −31.20 mV).

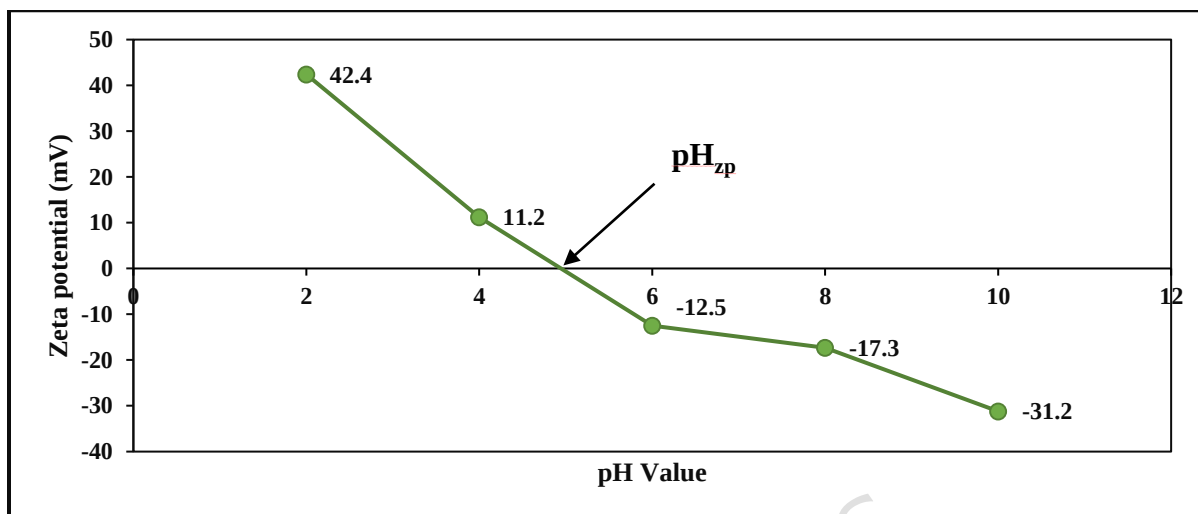


Fig. 10. Zeta potential results of Zr-MOF/CuCo₂O₄ nanocomposite.

Optimization of Synthesis Parameters

The RSM optimization results (Table 2) were crucial in identifying the optimal conditions for forming an active Zr-MOF/CuCo₂O₄ nanocomposite, particularly concerning the Cu:Co molar ratio. From a practical standpoint, both the 1:2 and 1.5:3 ratios yielded the highest tetracycline removal efficiencies (as detailed in Table 2), indicating a robust window for spinel CuCo₂O₄ formation and dispersion within the MOF structure. However, the 1:2 ratio is preferred as it achieves this optimal performance with a lower overall quantity of precursor salts, representing a more efficient and cost-effective synthesis approach. In contrast, the 0.5:1 ratio resulted in lower removal efficiencies, suggesting incomplete spinel formation and insufficient active sites.

This structural optimization directly explains the significantly enhanced tetracycline removal efficiency observed with the optimized ratios (1:2 and 1.5:3) compared to the individual components. As shown in Table 3, neither Zr-MOF (61.0%) nor CuCo₂O₄ alone (34.0%) achieves high adsorption capacity because

each component provides only part of the necessary interactions: Zr-MOF contributes porosity and Lewis acid sites, while CuCo_2O_4 provides redox-active metal centers. When combined in the optimized ratios, a synergistic effect emerges. The well-dispersed CuCo_2O_4 nanoparticles enhance the surface charge, increase the density of coordination sites, and facilitate π - π /electrostatic interactions with tetracycline molecules. Practically, the optimal RSM conditions correspond to a composite structure that maximizes accessible active sites and preserves effective diffusion pathways, leading to the significantly improved removal performance of the nanocomposite.

Zr-MOF (g)	$\text{Cu}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (mmol)	$\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (mmol)	Removal (%)
0.84	0.5	1	56.88
0.84	1.0	2	83.00
0.84	1.5	3	83.00

Table 2. Results of optimization of synthesis amount Zr-MOF and the synthesis precursors of CuCo_2O_4 .

Samples	Removal (%)
Zr-MOF	61.00
CuCo_2O_4	34.00
Zr-MOF/ CuCo_2O_4	>99

Table 3. Adsorption performance of Zr-MOF, CuCo_2O_4 and Zr-MOF/ CuCo_2O_4 for removal of TC.

Optimization of TC Removal Using RSM

In the present research, the RSM based on Central Composite Design (CCD), utilizing Design-Expert software (v13), was employed to model and analyze the effects of four independent variables: (A) pH, (B) initial TC concentration, (C) nanoadsorbent amount, and (D) contact time. The experimental design

included one replicate, five levels, two blocks, and six center points, with one response (removal efficiency %). A total of 30 runs were conducted. Thirty experiments were designed and performed based on the coded levels of the variables, leading to the development of a quadratic regression model (Eq. 2):

$$[Y(\%) = 33.6208 + 12.1842A - 13.6892B + 12.7458C + 4.15917D - 11.075AB - 1.995AC + 6.95125AD + 5.1BC - 2.43625BD - 4.38125CD + 0.691458A^2 + 8.72396B^2 + 2.67396C^2 + 7.24271D^2] \quad (2)$$

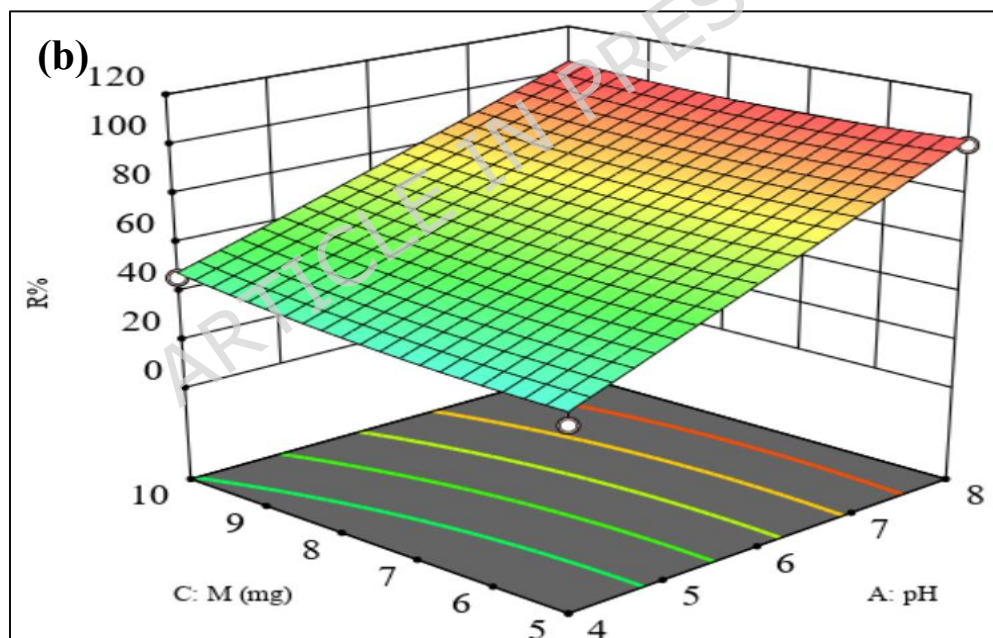
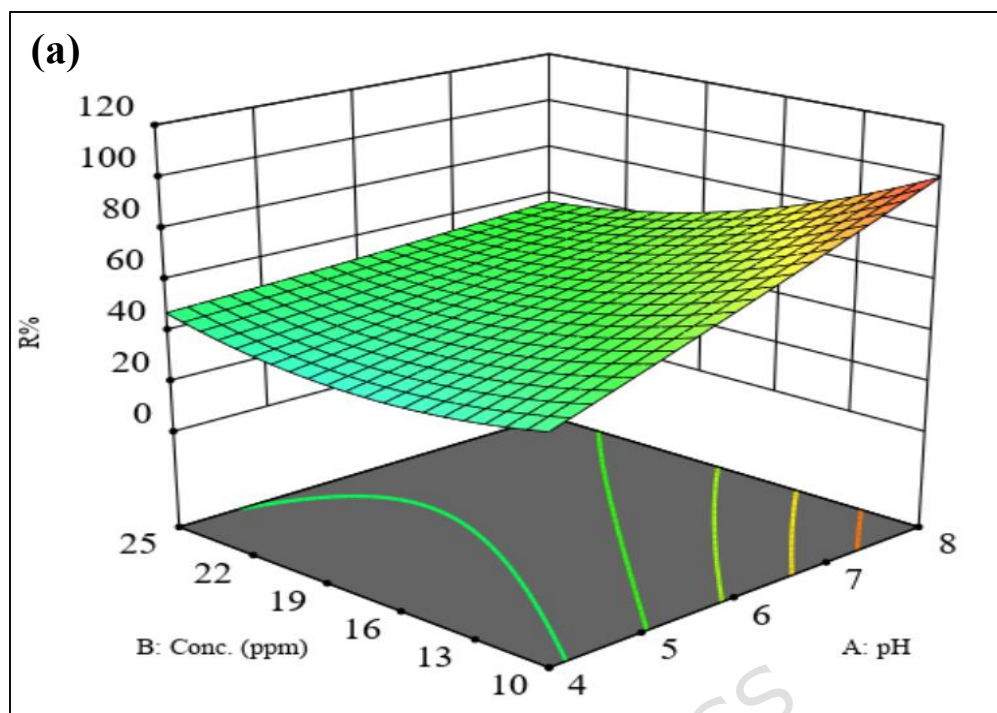
ANOVA analysis confirmed the model's significance, with an R^2 of 0.9919 and an adjusted R^2 of 0.9837, indicating excellent agreement between predicted and experimental data. The predicted R^2 of 0.9822 further supports the model's validity (model significance p-value < 0.0001). The lack of fit was found to be non-significant (p-value = 0.9896), which is desirable. Independent parameters, including main effects, two-factor interactions, and quadratic terms (A, B, C, D, A^2 , B^2 , C^2 , D^2 , AB, AC, AD, BC, BD, CD), with p-values less than 0.05, were identified as significant model terms for TC removal. The adequate precision value of 25.40 confirmed a strong signal-to-noise ratio.

Three-dimensional response surfaces (Fig. 11) revealed that the removal percentage improved with increasing adsorbent content and pH (A). Conversely, it decreased with increasing initial TC concentration (B), attributed to saturation effects and potential turbidity^{39,40}. Specifically, as the initial TC concentration increased, the removal efficiency decreased due to the limited number of active sites available relative to the increased TC molecules. The optimal conditions were determined as: pH 7.80, 17.50 mgL⁻¹ initial TC concentration, 7.0 mg adsorbent, and 46 min contact time, achieving a predicted >99% TC removal. The optimized coordinates for the adsorption process (see Table 4) were obtained via optimization using Design

Expert software based on the RSM. To validate the RSM prediction accuracy, three experiments were conducted at the predicted optimum coordinates. As presented in Table 4, the experimental results closely matched the predicted responses.

pH	Concentration TC (ppm)	Adsorbent dosage (mg)	Time (min)	Removal efficiency (%)	
				From model	Experimental
7.80	17.50	7.0	46	99.94	99.87±0.69%

Table 4. Model validation and optimum point.



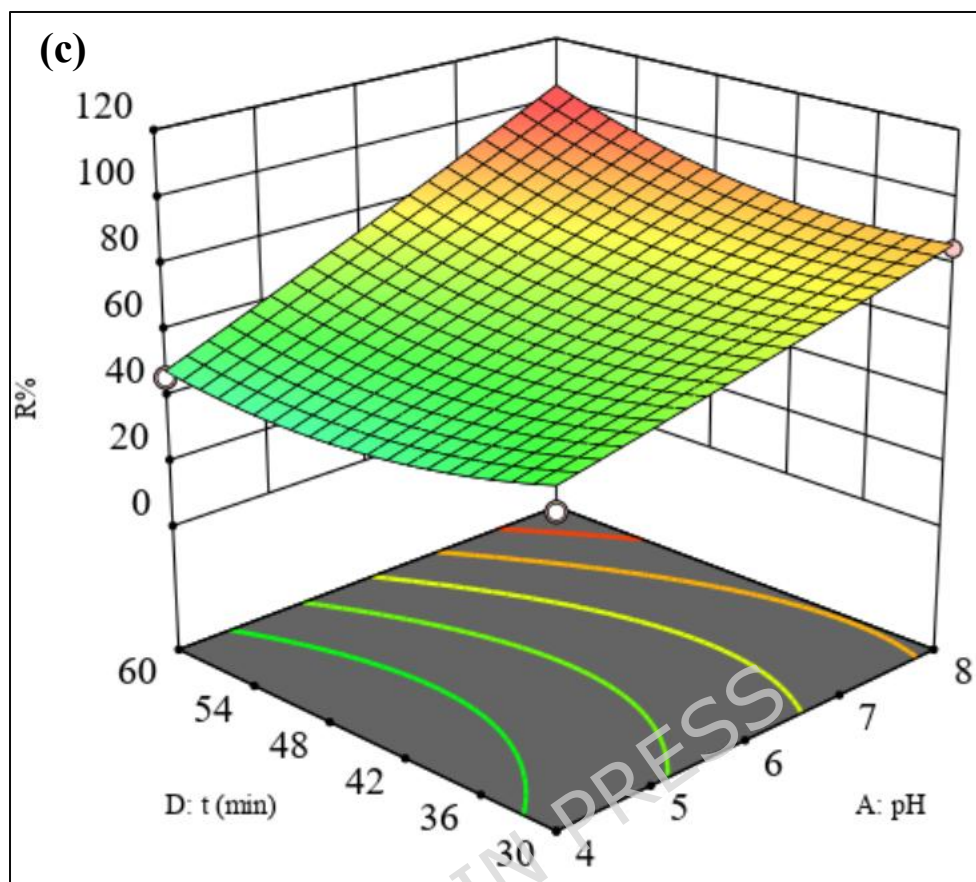
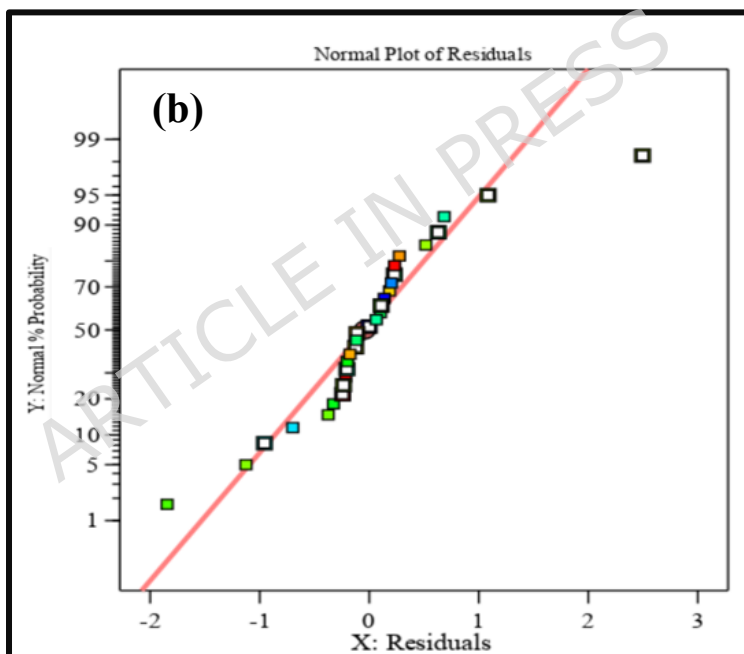
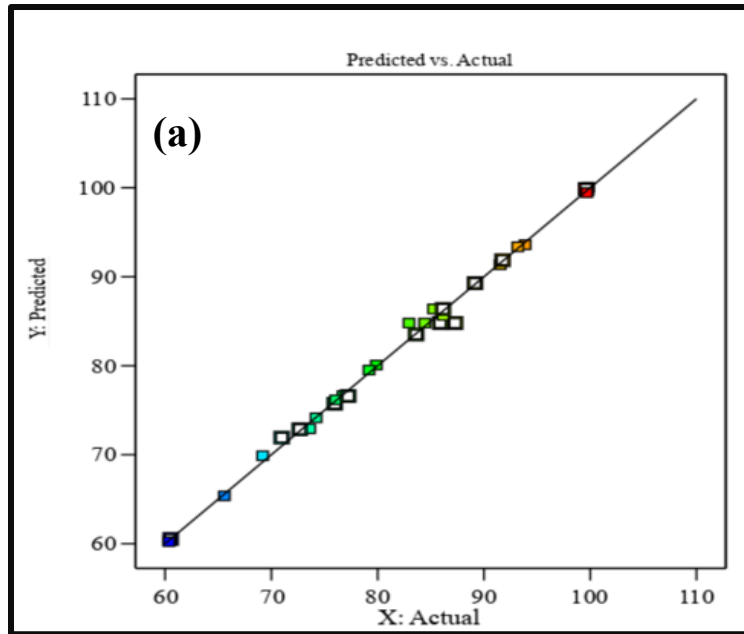


Fig. 11. Three-dimensional graphs of experimental design results.

The diagnostic plots (Fig. 12) provide strong evidence for the validity of the regression model. The comparison between actual and predicted values (Fig. 12a) demonstrates the excellent accuracy of TC removal efficiency prediction by the Zr-MOF/CuCo₂O₄ nanocomposite. Further validation through residual analysis, where residuals were observed to be randomly and uniformly distributed, indicates that the model generalizes well and overfitting is not a significant concern with the current parameterization. The residual analyses (Fig. 12b–d) reveal a normal distribution and random behavior of residuals, confirming that the fundamental assumptions of regression are satisfied. Overall, these data confirm the robustness and reliability of the presented model in explaining the experimental findings.



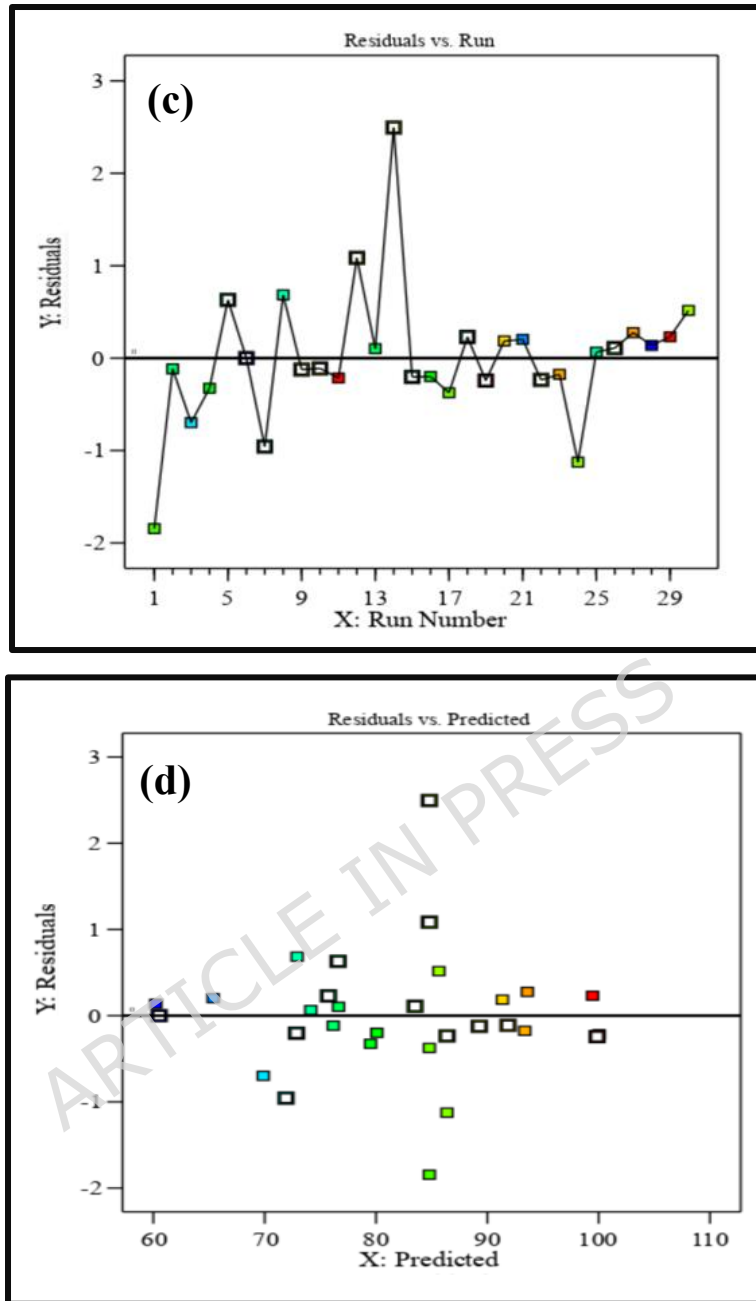


Fig. 12. Plots of the residuals 30 runs for TC removal by Zr-MOF/CuCo₂O₄.

Evaluation of the Method

The adsorption equilibrium for TC removal was described using Eq. (3):

$$q_e = \frac{(C_0 - C_e) \times V}{M} \quad (3)$$

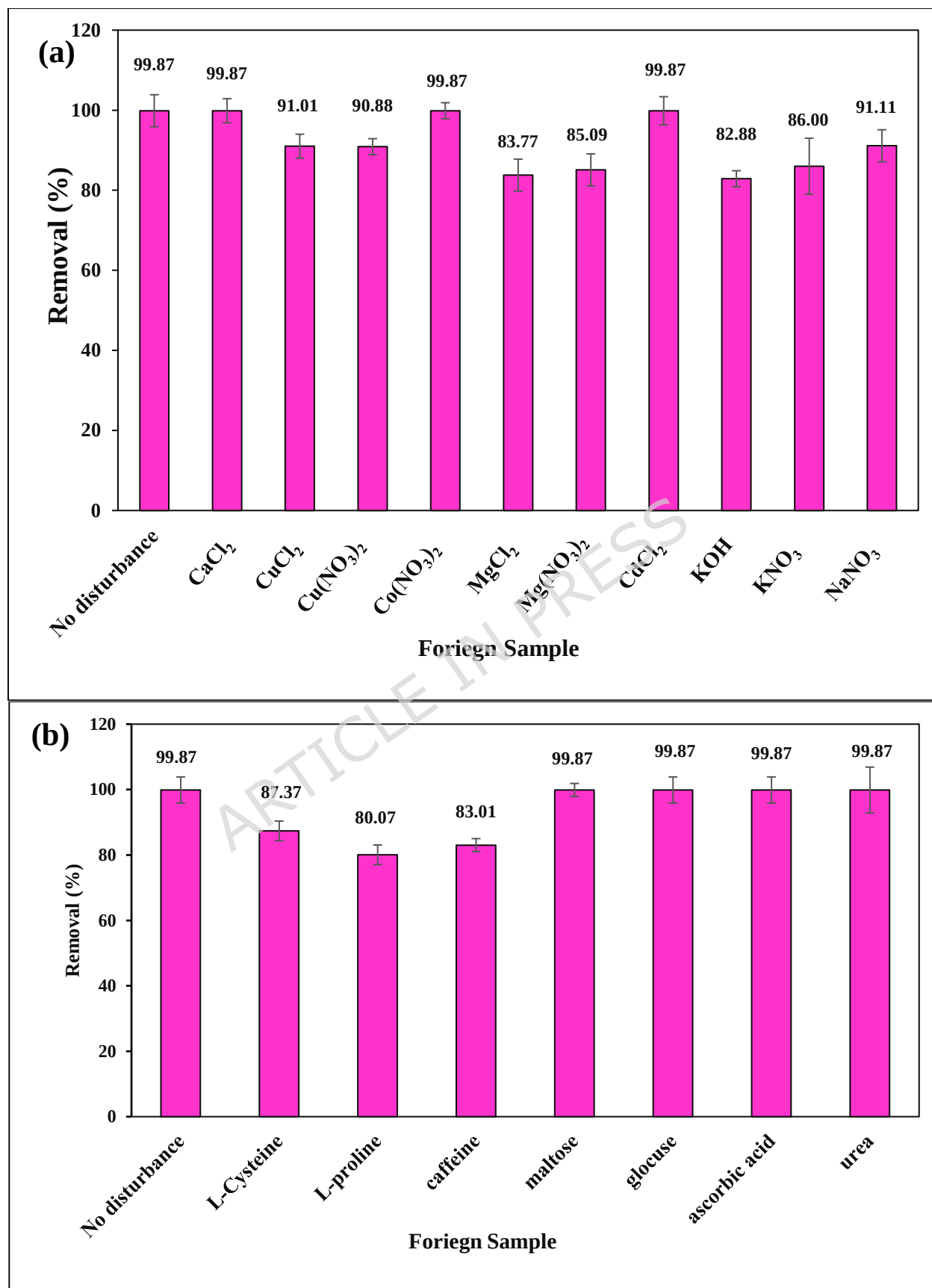
where C_0 and C_e (mgL^{-1}) represent the initial and equilibrium TC concentrations, V (L) is the solution volume, and M (g) is the adsorbent mass⁴¹. The Zr-MOF/ CuCo_2O_4 nanoadsorbent, with a maximum adsorption capacity of 100 mgg^{-1} , shows very high efficiency in adsorbing TC antibiotic molecules⁴².

Effect of Coexisting Ions and Organic Compounds

To evaluate selectivity, TC removal was studied in the presence of common inorganic salts such as CaCl_2 , CuCl_2 , $\text{Cu}(\text{NO}_3)_2$, $\text{Co}(\text{NO}_3)_2$, MgCl_2 , $\text{Mg}(\text{NO}_3)_2$, $\text{CdCl}_2 \cdot 2\text{H}_2\text{O}$, KOH , KNO_3 , and NaNO_3 (1 mM each). Under optimal conditions, the amount of each coexisting salt was added. Removal efficiency decreased by $>5\%$ in the presence of Cu^{2+} , Mg^{2+} , K^+ , and Na^+ salts, attributed to competition for active adsorption sites and alteration of surface charge (Fig. 13a).

Organic compounds such as L-cysteine, L-proline, caffeine, maltose, glucose, ascorbic acid, and urea (1 mM) were also tested. Cysteine, L-proline, and caffeine significantly interfered with TC adsorption (Fig. 13b).

The influence of other antibiotics (azithromycin, sulfamethoxazole, cefazolin, trimethoprim, ofloxacin, cefixime, and metronidazole) was examined (1 mM each). Ofloxacin, cefixime, and metronidazole interfered with TC removal (Fig. 13c).



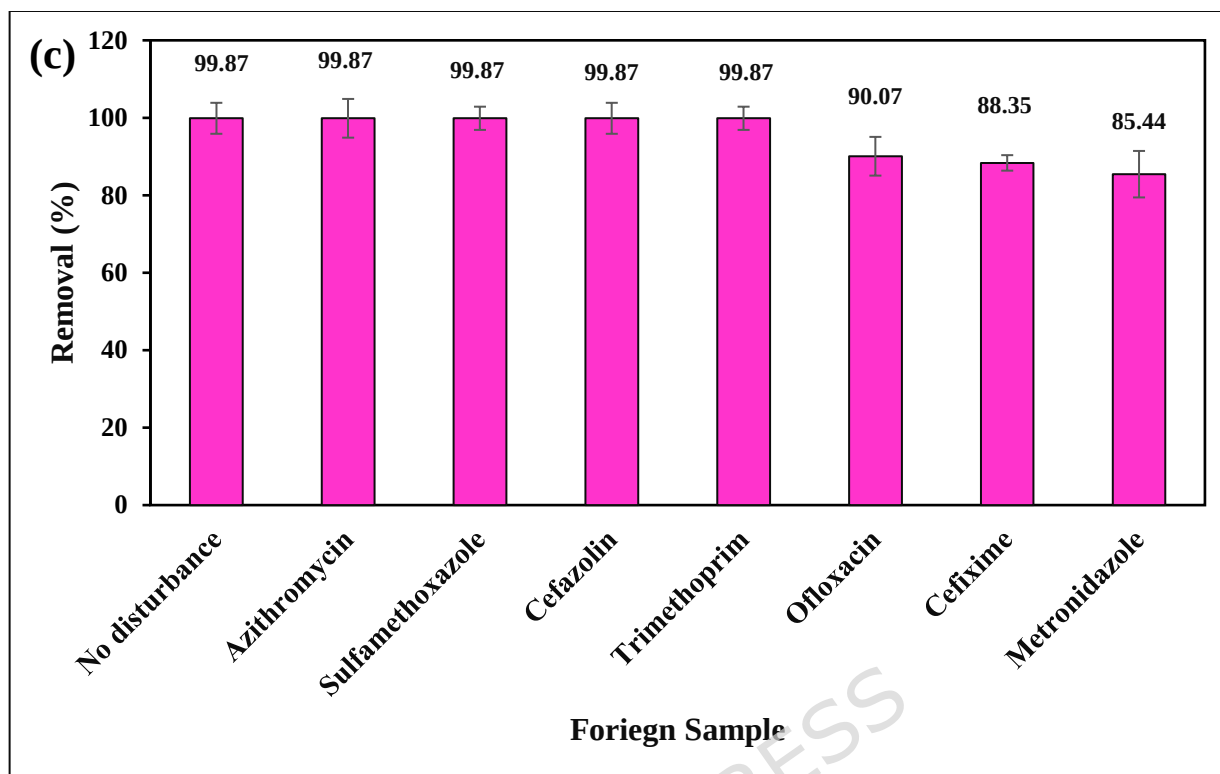


Fig. 13. Coexisting samples test of Zr-MOF/CuCo₂O₄ nanocomposite for TC removal.

Real Sample Analysis

The adsorbent performance was validated in real samples. To assess the adsorbent's performance under realistic conditions, seawater from the Persian Gulf (Qeshm Island) was utilized. Given that the salinity of the Persian Gulf typically falls between 3600-4800 mg/L, this represents a saline environment where the adsorbent demonstrated a high tetracycline removal efficiency of 98.0%. In distilled water, >99% removal was observed.

Additionally, TC removal was tested using pharmaceutical-grade tetracycline (400 mg tetracycline, Sina Darou, Iran). A solution with optimal concentration was prepared and tested under optimal conditions. The composite removed 98.43% of TC, demonstrating strong applicability in real-world systems.

Adsorption Reusability and Stability

Desorption tests were conducted using various organic solvents, including DMSO, ethanol, methanol, and acetonitrile. Ethanol demonstrated the highest desorption efficiency among the tested solvents. The reusability of the adsorbent was subsequently evaluated over seven consecutive cycles. The removal efficiency exhibited a gradual decrease from >99% in the first cycle to 85.05% in the seventh cycle. This decline was primarily attributed to incomplete regeneration and potential site blockage. Factors contributing to these decreases in capacity and regeneration efficiency include the reduced availability of active adsorption sites and compromised porosity, likely caused by incomplete desorption of contaminants from the adsorbent surface. Furthermore, interactions between the adsorbent and TC may have weakened over the cycles. These findings collectively demonstrate the considerable structural stability and reusability of the Zr-MOF/CuCo₂O₄ nanocomposite for multi-cycle operations. To ascertain the presence and nature of functional groups and specific bands within the nanocomposite, a comparative analysis of FT-IR spectra was performed before and after the stability test (Fig. 14). The FT-IR spectra indicated only minor changes in the adsorbent's structure, confirming its overall robustness.

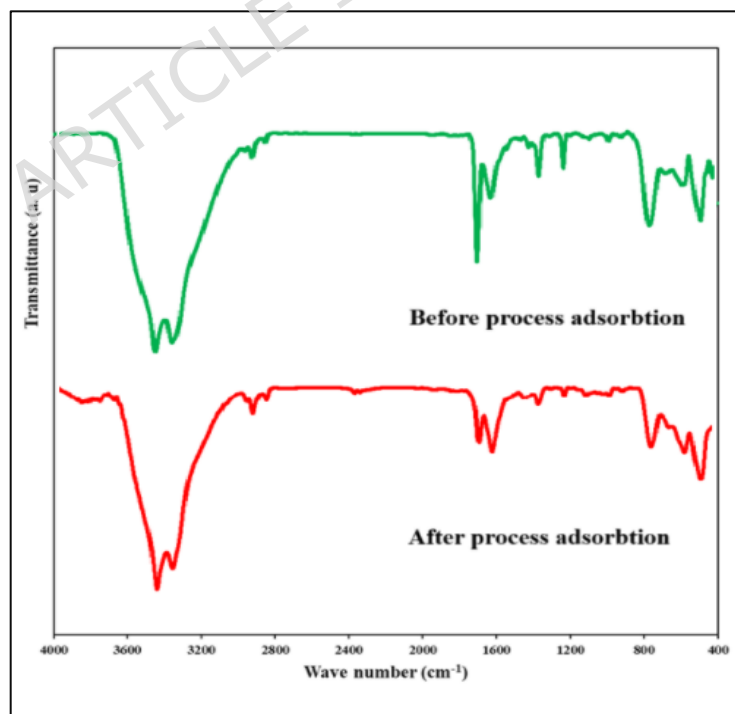


Fig. 14. FT-IR spectra before and after process adsorption TC of Zr-MOF/CuCo₂O₄.**Adsorption Isotherms, Kinetics, and Thermodynamics****Adsorption Isotherms**

Equilibrium adsorption data were fitted to various isotherm models⁴³. The parameters for each isotherm equation are given in the referenced paper⁴⁴. Experiments were performed at 300 K using 7.0 mg of adsorbent and TC concentrations (50-400 mgL⁻¹).

The result fit the Langmuir model ($R^2 = 0.99$) indicating that the adsorption process occurred in a monolayer and on homogeneous sites. Comparing the statistical parameters show that the equilibrium data of TC adsorption align well with the Langmuir model. Considering the comparison of equilibrium data of the TC adsorption using Zr-MOF/CuCo₂O₄ with the Langmuir model and the assumptions of this model, it can be concluded that the adsorption occurs in a single layer. This suggests that each active site on Zr-MOF/CuCo₂O₄ interacts with one TC molecule, confirming chemisorption dominance (Table 5).

Isotherm	Equation	Parameters	Values for parameters
Langmuir	$C_e/Q_e = 1/bQ_0 + C_e/Q_0$	Q_e (mgg ⁻¹)	15.33±0.41%
		R^2	0.99±0.39%
Freundlich	$\log q_e = \log K_F + (1/n) \log C_e$	K_F (mgg ⁻¹)	20.44±0.51%
		R^2	0.98±0.56%
Temkin	$q_e = R_T/b_T \ln A_T + (R_T/b_T) \ln C_e$	K_T (mgL ⁻¹)	12.73±0.46%
		R^2	0.95±0.49%
Dubinin-Radashkuvich	$\ln(q_e) = \ln(q_s) - k_{ad}\epsilon^2$	Q_s (mgg ⁻¹)	18.12±0.34%
		R^2	0.93±0.38%

Table. 5. Isotherms of TC adsorption on the Zr-MOF/CuCo₂O₄ adsorbent (10.0 mL solution, 0.70 gL⁻¹ sorbent dosage, pH =7.80 and T = 300 K).

Adsorption Kinetics

To elucidate adsorption dynamics, kinetic models were evaluated⁴⁵⁻⁴⁷. The adsorption followed pseudo-second-order kinetics, described by Eq. (4):

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \quad (4)$$

where q_t and q_e (mgg⁻¹) denote adsorption capacities at time t and equilibrium, respectively, and k_2 (gmg⁻¹min⁻¹) is the rate constant. The obtained parameters ($k_2 = 0.09 \text{ gmg}^{-1}\text{min}^{-1}$, $R^2 = 0.9999$) confirm a strong correlation with experimental data, supporting a chemisorption mechanism (Fig. 15)⁴⁸⁻⁵⁰. After deeply understand the kinetic process of TC absorption on Zr-MOF/CuCo₂O₄, pseudo-second-order was used to fit the obtained experimental data linearly. Figure 16 presents the main adsorption mechanism for TC molecules adsorption using Zr-MOF/CuCo₂O₄ adsorbent⁵¹⁻⁵². Various chemical interactions, including surface complexation, hydrogen bonding, Lewis acid base interactions, π - π interactions, and physical adsorption, contributed to the adsorption of the TC molecules by Zr-MOF/CuCo₂O₄ adsorbent^{53,54}.

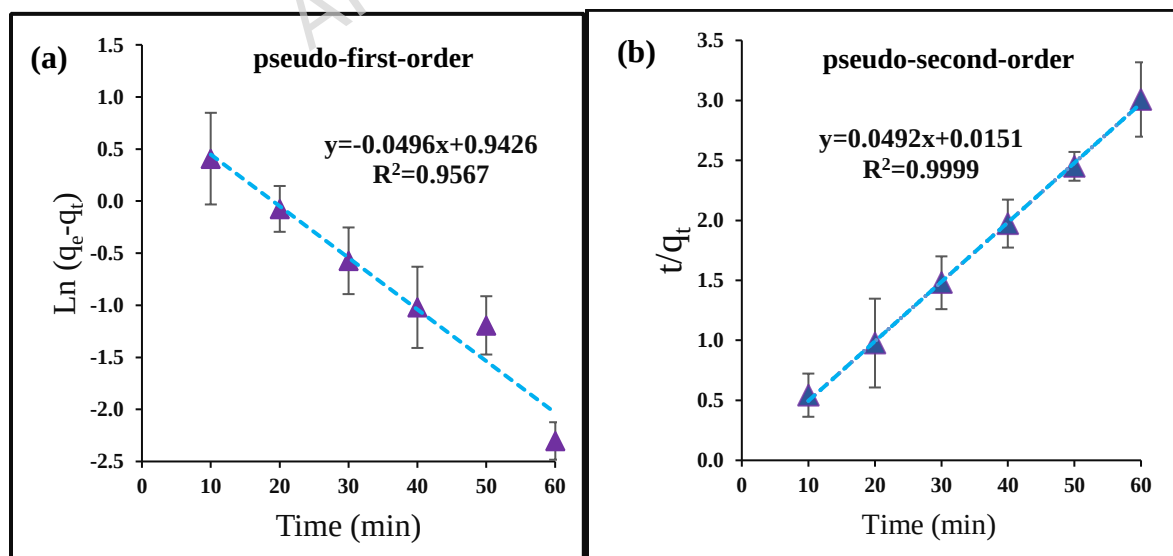


Fig. 15. Results of a kinetic study on Zr-MOF/CuCo₂O₄.

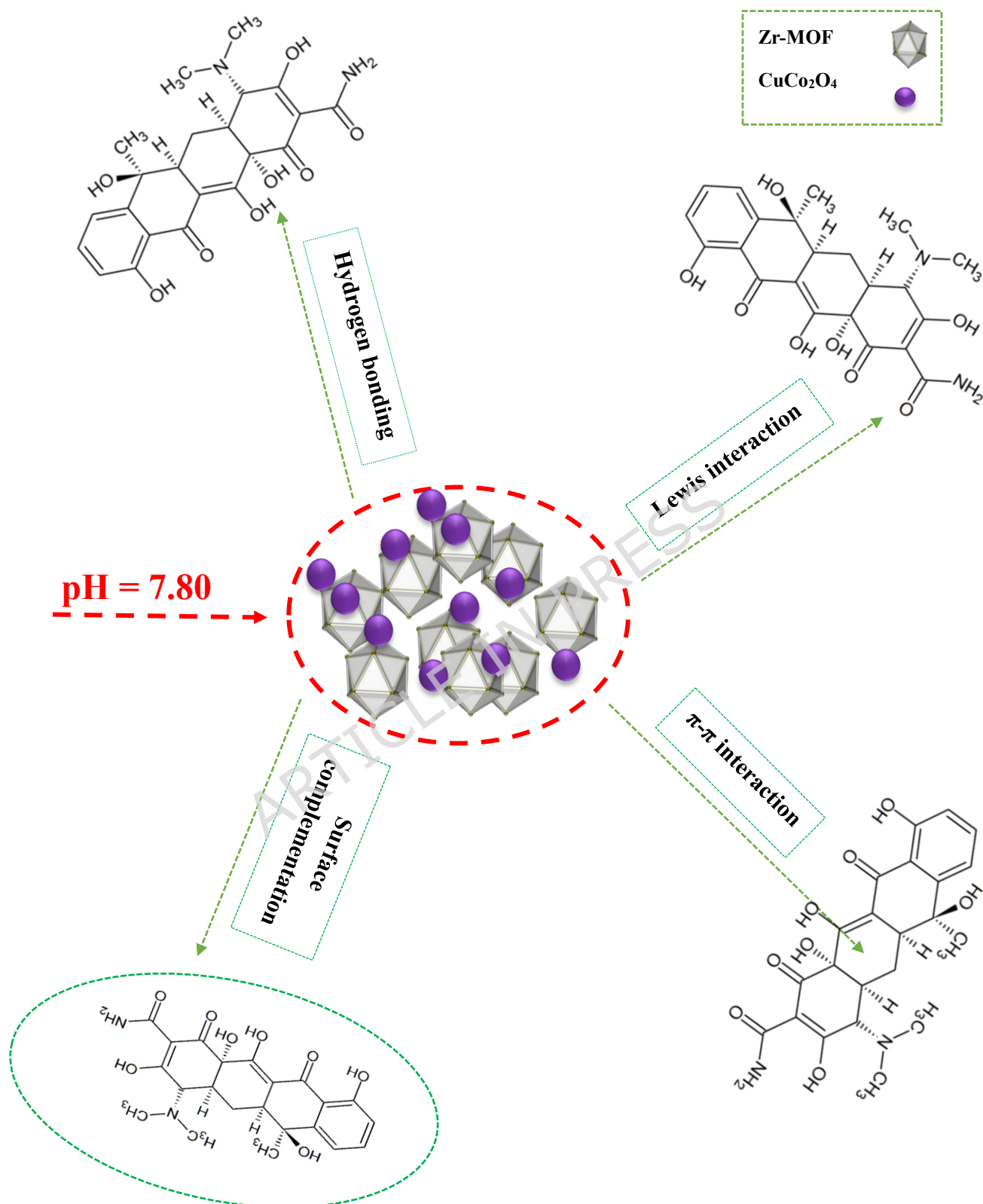


Fig. 16. Adsorption possible mechanism TC on Zr-MOF/CuCo₂O₄.

Thermodynamic Analysis

The thermodynamic parameters were calculated using the van't Hoff equation Eq. (5)⁵⁵:

$$\ln K_c = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{RT} \quad (5)$$

where $K_c = q_e/C_e$, R is the gas constant, and T (K) is temperature. Calculated values of Gibbs free energy (ΔG°), enthalpy (ΔH°), and entropy (ΔS°) are summarized in Table 6.

ΔG° values were -3.20 , -3.54 , -3.62 , -4.16 , and -4.37 kJmol^{-1} at 308–353 K, respectively, indicating spontaneous adsorption. The positive ΔH° (17.11 kJmol^{-1}), confirmed an endothermic process, requiring some energy input, consistent with moderate interaction strengths. while the positive ΔS° (33.15 $\text{J mol}^{-1}\text{K}^{-1}$) reflected increased randomness and affinity between TC and Zr-MOF/CuCo₂O₄, suggesting increased disorder upon adsorption, possibly due to the release of solvent molecules. This combination of moderate endothermicity and increased randomness points towards an adsorption mechanism that involves more than just weak van der Waals forces, potentially including initial bond formation or surface rearrangement characteristic of chemisorption, albeit not as strong as in some other systems^{53, 56-58}.

Adsorbate	ΔH° (kJmol ⁻¹)	ΔS° (Jmol ⁻¹ K ⁻¹)	ΔG°				
			308 K	323 K	333 K	343 K	353 K
TC	17.11	33.15	-3.20	-3.54	-3.62	-4.16	-4.37

Table 6. Results related to the thermodynamic study.

Comparative Performance Analysis

Table 7 compares the TC removal performance of Zr-MOF/CuCo₂O₄ with other reported adsorbents⁵⁹⁻⁶². The results indicate that the present nanocomposite exhibits comparable or superior removal efficiency and reusability, underscoring its potential as an advanced material for wastewater remediation.

Notably, this novel nanoadsorbent demonstrates better and comparable performance to previously reported materials in several aspects. For instance, it achieves a more favorable time ratio compared to adsorbents such as Al-UiO-66-NH₂, Ni-UiO-66, and UiO-66-(COOH)₂/GO. Furthermore, our adsorbent exhibits a higher removal efficiency relative to TC compared to Zr-MOF@Fe₃O₄ and Al-UiO-66-NH₂. The optimal pH for our material was found to be 7.80, which is notably different from the optimal pH values reported for other adsorbents (e.g., pH 4, 11, and 3).

No.	Adsorbent	Pollutant	dosage (g/L)	Time (min)	pH	Temperature (K)	Removal (%)	References
1	Zr-MOF@Fe ₃ O ₄	tetracycline	10	45	4	300	91.22	59
2	Al-UiO-66-NH ₂	tetracycline	0.2	70	7	300	85.27	60
3	Ni-UiO-66	tetracycline	20	60	11	300	100	61
4	UiO-66- (COOH) ₂ /GO	tetracycline	0.5	2880	3	300	-	62
5	Zr-MOF/ CuCo ₂ O ₄	tetracycline	0.70	46	7.80	300	99.87	This work

Table 7. Compares the adsorbent studied in this article with other adsorbents.

Conclusion

The successful synthesis and application of the Zr-MOF/CuCo₂O₄ nanocomposite for tetracycline removal demonstrates its strong environmental significance. By achieving >99% removal efficiency under optimized conditions and maintaining stability over multiple reuse cycles, this material offers a sustainable solution for mitigating antibiotic pollution in aquatic environments. While the positive enthalpy change ($\Delta H = +17.11 \text{ kJmol}^{-1}$) indicates an endothermic process, which can be associated with chemisorption, its relatively low magnitude, coupled with a positive entropy change ($\Delta S = +33.15 \text{ Jmol}^{-1}\text{K}^{-1}$), warrants careful interpretation. This combination of thermodynamic parameters suggests that the adsorption might involve a moderate interaction strength, potentially bridging physisorption and weaker forms of chemisorption, rather than very strong, irreversible chemical bond formation. The positive ΔS further supports the notion of increased molecular freedom upon adsorption. The ability to perform effectively in real samples such as

seawater and pharmaceutical wastewater highlights its practical applicability for large-scale water treatment systems. Based on the data obtained, the efficiency of this nanocomposite is not limited to its extraordinary adsorption capacity and rate, but also acts to reduce the ecological risks of antibiotic contaminants, maintain the quality of water resources, and comply with environmental protection regulations. These observations establish Zr-MOF/CuCo₂O₄ as a promising candidate for application in next-generation water purification and pollution control technologies.

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Authorship contributions

Shahrbab Dadari Doolabi: Data curation, Investigation, Formal analysis, Writing – original draft. Tayebbeh Shamspur: Project administration, Review & editing, Methodology, Supervision. Ali Mostafavi: Project administration, Supervision. Maryam Mohamadi: Data curation, Review & editing.

Data Availability

Data from this research are available upon request from the corresponding author.

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Declarations**Competing interests**

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